Bank Transaction Analysis

📌 Objective

The objective of this project is to design and analyze a banking database system using SQL. The project focuses on understanding customer transactions, account details, and financial trends through structured data analysis.

📌 Tools & Technologies

SQL (Structured Query Language)

Relational Database Management System

📌 Database Structure

The project consists of three main tables:

1. Customers

Stores customer details such as customer ID, name, and city.

2. Accounts

Contains account-related information including account type and balance, linked to customers.

3. Transactions

Stores transaction records such as deposits and withdrawals along with transaction dates.

📌 Key Features

- Creation of relational databases with multiple tables
- Data insertion and management using SQL
- Use of JOIN operations to connect multiple tables
- Application of GROUP BY and aggregation functions
- Analysis of financial transactions and customer behavior

📊 Analysis Performed

- Retrieved customer transaction details using JOIN operations
- Calculated total balance for each customer
- Compared total deposits and withdrawals
- Identified customers with highest account balance
- Analyzed monthly transaction trends

📈 Key Insights

- Identified transaction patterns across different customers
- Observed differences between deposit and withdrawal amounts
- Recognized high-value customers based on account balance
- Understood monthly transaction flow

🎯 Conclusion

This project demonstrates the ability to design a relational database and perform meaningful data analysis using SQL. It highlights practical skills in handling structured data, writing efficient queries, and extracting business insights from raw data.